

FOUNDATIONS OF DESCRIPTIVE STATISTICAL ANALYSIS

Master Solutions Guide: Tutorials 1, 2, 3, 4, and 5 (Complete 7 Questions)

Comprehensive Solutions Reference: This document compiles all questions, problem statements, and step-by-step mathematical solutions across **Tutorials 1, 2, 3, 4, and 5**. Tutorial 5 specifically contains all **7 complete questions and answers** as requested.

Tutorial 1: Probability Fundamentals & Set Theory

Q1. Event Identities & Set Proofs

Prove mathematically using set operations and the axioms of probability:

- (a) $P(A) = P(A \cap B) + P(A \cap B')$
- (b) $P(A \cup B) = P(A \cap B) + P(A \cap B') + P(A' \cap B)$

Detailed Solution:

(a) Any event A can be partitioned into two disjoint sets: $A = (A \cap B) \cup (A \cap B')$. By Axiom 3 of probability (additivity for disjoint sets): **$P(A) = P(A \cap B) + P(A \cap B')$** .

(b) The union $A \cup B$ can be decomposed into three mutually exclusive regions: $(A \cap B) \cup (A \cap B') \cup (A' \cap B)$. Applying Axiom 3 yields: **$P(A \cup B) = P(A \cap B) + P(A \cap B') + P(A' \cap B)$** .

Q2. Complement Relation

Given $P(A \cap B) = p$ and $P(A \cup B) = q$, derive the exact expression for $P(A') + P(B')$.

Detailed Solution:

By Inclusion-Exclusion: $P(A \cup B) = P(A) + P(B) - P(A \cap B) \Rightarrow P(A) + P(B) = q + p$.

Using the complement rule: $P(A') = 1 - P(A)$ and $P(B') = 1 - P(B)$.

Summing complements: $P(A') + P(B') = (1 - P(A)) + (1 - P(B)) = 2 - (P(A) + P(B)) = 2 - p - q$.

Q3. Two Rolled Dice Probabilities

Two fair 6-sided dice are rolled simultaneously (36 sample outcomes). Find the probability that the sum is: (i) equal to 8, (ii) equal to 13, (iii) less than or equal to 12.

Detailed Solution:

- **(i) Sum = 8:** Favorable outcomes = $\{(2,6), (3,5), (4,4), (5,3), (6,2)\}$ (5 outcomes). $P(\text{Sum} = 8) = 5/36 \approx 0.1389$.
- **(ii) Sum = 13:** Maximum possible sum with two dice is 12. $P(\text{Sum} = 13) = 0$.
- **(iii) Sum ≤ 12 :** All outcomes have a sum ≤ 12 . $P(\text{Sum} \leq 12) = 36/36 = 1$.

Q4. Birthday Problem Complement

In a group of 3 students, the probability that 2 students do not have the same birthday is 0.992. What is the probability that 2 students have the same birthday?

Detailed Solution:

$P(\text{Same Birthday}) = 1 - P(\text{Not Same Birthday}) = 1 - 0.992 = 0.008$ (0.8%).

Q5. Marble Selection Probabilities

A box contains 3 blue, 2 white, and 4 red marbles (total = 9). A marble is drawn at random. Find the probability that it is: (i) white, (ii) blue, (iii) red.

Detailed Solution:

Total = 3 + 2 + 4 = 9.

- (i) $P(\text{White}) = 2/9 \approx 0.2222$
- (ii) $P(\text{Blue}) = 3/9 = 1/3 \approx 0.3333$
- (iii) $P(\text{Red}) = 4/9 \approx 0.4444$

Q6. Ball Selection

A bag contains 3 red balls and 5 black balls (total = 8). A ball is drawn at random. Find the probability that it is: (i) red, (ii) not red.

Detailed Solution:

- (i) $P(\text{Red}) = 3/8 = 0.375$
- (ii) $P(\text{Not Red}) = 5/8 = 0.625$

Q7. Group AB Blood Sampling Probability

$P(\text{Group AB}) = 0.0526$. If 20 subjects are randomly selected, find the probability of getting at least one with Group AB blood.

Detailed Solution:

$P(\text{None AB}) = (1 - 0.0526)^{20} = (0.9474)^{20} \approx 0.3396$.

$P(\text{At least 1 AB}) = 1 - 0.3396 = 0.6604$ (66.04%). Not high enough for clinical certainty.

Tutorial 2: Conditional Probability, Diagnostic Screening & Blood Types

Q1. Cancer Diagnostic Screening (Bayes' Theorem)

Prevalence rate = 1% ($P(C) = 0.01$). Sensitivity (True Positive) = 80% ($P(+|C) = 0.80$). False Positive = 10% ($P(+|C') = 0.10$). Calculate $P(C | +)$, the probability a person has cancer given a positive result.

Detailed Solution:

$$P(C | +) = [P(+ | C) P(C)] / [P(+ | C) P(C) + P(+ | C') P(C')]$$

$$P(C | +) = (0.80 \times 0.01) / [(0.80 \times 0.01) + (0.10 \times 0.99)] = 0.008 / (0.008 + 0.099) = 0.008 / 0.107 \approx 0.0748 \text{ (7.48\%)}$$

Q2. Pre-Employment Drug Screening Analysis

A study of 555 drug test subjects yields the following contingency table:

- Subject Uses Drugs: 45 Positive, 5 Negative (Total = 50)
- Subject Does Not Use Drugs: 25 Positive, 480 Negative (Total = 505)

Find: (a) $P(\text{Positive} | \text{Uses Drugs})$, (b) $P(\text{Uses Drugs} | \text{Positive})$.

Category	Positive Test Result	Negative Test Result	Total
Subject Uses Drugs	45 (True Positive)	5 (False Negative)	50
Subject Does Not Use Drugs	25 (False Positive)	480 (True Negative)	505
Total	70	485	555

Detailed Solution:

(a) $P(\text{Positive} | \text{Uses Drugs})$: Restrict to 50 drug users: $45 / 50 = 0.900$ (90.0%).

(b) $P(\text{Uses Drugs} | \text{Positive})$: Restrict to 70 positive test results: $45 / 70 \approx 0.6429$ (64.3%).

Q3. Blood Groups & Rh Types (Sampling with vs. without Replacement)

Evaluate independence when selecting two people for blood characteristics under sampling with vs. without replacement.

Key Concepts & Solutions:

- **With Replacement:** Events are strictly independent because the population distribution remains unchanged.
- **Without Replacement:** Events are dependent. However, by the 5% Guideline, if sample $n \leq 5\%$ of population N , selections can be treated as independent for calculation purposes.

Tutorial 3: Continuous Distributions & Normal Curve Analysis

Q4. Gamma Probability Distribution Parameters

Find the mean μ and variance σ^2 for a Gamma distribution with:

- (a) $\alpha = 2, \beta = 3$
- (b) $\alpha = 3, \beta = 4$

Detailed Solution:

Gamma Formulas: $\mu = \alpha\beta$ and $\sigma^2 = \alpha\beta^2$.

(a) $\alpha = 2, \beta = 3$: $\mu = 2 \times 3 = 6$; $\sigma^2 = 2 \times 3^2 = 18$.

(b) $\alpha = 3, \beta = 4$: $\mu = 3 \times 4 = 12$; $\sigma^2 = 3 \times 4^2 = 48$.

Q5. Standard Normal Curve Critical z-Scores

Find z for the standard normal curve area:

- (a) between 0 and z is 0.4726
- (b) to the left of z is 0.9868
- (c) to the right of z is 0.1314
- (d) between $-z$ and z is 0.8502

Detailed Solution:

(a) Cumulative left area = $0.5 + 0.4726 = 0.9726 \Rightarrow z = 1.92$.

(b) Area = 0.9868 $\Rightarrow z = 2.22$.

(c) Left area = $1 - 0.1314 = 0.8686 \Rightarrow z = 1.12$.

(d) Tail area = $(1 - 0.8502)/2 = 0.0749 \Rightarrow$ Left area = 0.9251 $\Rightarrow z = 1.44$.

Q6. Normal Distribution Probability Computations

For $X \sim N(\mu = 10, \sigma^2 = 36)$ ($\sigma = 6$), calculate:

- (a) $P(X > 5)$, (b) $P(4 < X < 16)$, (c) $P(X < 8)$, (d) $P(X < 20)$, (e) $P(X > 16)$.

Detailed Solution (standardizing $z = (x - 10)/6$):

(a) $P(X > 5)$: $z = (5-10)/6 = -0.83 \Rightarrow P(Z > -0.83) = 1 - 0.2033 = 0.7967$.

(b) $P(4 < X < 16)$: $z_1 = -1.00, z_2 = +1.00 \Rightarrow P(-1 < Z < 1) = 0.8413 - 0.1587 = 0.6826$.

(c) $P(X < 8)$: $z = (8-10)/6 = -0.33 \Rightarrow P(Z < -0.33) = 0.3707$.

(d) $P(X < 20)$: $z = (20-10)/6 = +1.67 \Rightarrow P(Z < 1.67) = 0.9525$.

(e) $P(X > 16)$: $z = +1.00 \Rightarrow P(Z > 1.00) = 0.1587$.

Tutorial 4: Frequency Tables, Measures of Center & Visualizations

Problem 1. Frequency Table & Bar Chart for Coin Tosses

A 3-coin toss experiment repeated 30 times recorded the number of heads.

Solution: Tally counts for $x \in \{0, 1, 2, 3\}$ are tabulated to form frequency f and relative frequency $rf = f / 30$, displayed via a discrete bar chart.

Problem 2. Grouped Frequency Table (Car Battery Lifespans)

Group 40 car battery lifespans into a frequency table with width 0.5 starting at 2.0.

Solution: Classes: 2.0–2.4, 2.5–2.9, 3.0–3.4, 3.5–3.9, 4.0–4.4, 4.5–4.9.

Class midpoints (2.2, 2.7, 3.2, 3.7, 4.2, 4.7) serve as horizontal coordinates for the histogram and frequency polygon.

Problem 3. Mean & Median Salary from Grouped Data

Compute mean and median for 60 employees grouped by salary bracket.

Solution Formulas:

- $\text{Mean } \bar{x} = \sum(f \cdot x_m) / n$
- $\text{Median} = L + [(n/2 - CF_{\text{prev}}) / f_{\text{median}}] \cdot w$

Problem 4. Central Tendency & Survey Analysis

Determine the mode for discrete survey categorical data and analyze distribution skewness.

Solution: Mode = response with highest frequency. Right skewness occurs when mean > median.

Tutorial 5: Summary Statistics, Quantiles & Comprehensive Analysis (All 7 Questions)

COMPLETE 7-QUESTION MASTER SECTION: All 7 questions from Tutorial 5 are comprehensively detailed below with step-by-step mathematical calculations, formulas, and analytical interpretations.

Question 1. Sample Data Summary Statistics

For the sample dataset: {120, 122, 125, 127, 128, 130, 132, 135} ($n = 8$), calculate:

(a) Sample Mean $\bar{x}$, (b) Sample Median, (c) Mode, (d) Quartiles $Q1, Q2, Q3$, (e) Interquartile Range (IQR).

Step-by-Step Solution:

(a) Sample Mean $\bar{x}$: $\sum x = 120 + 122 + 125 + 127 + 128 + 130 + 132 + 135 = 1019$.

$\bar{x} = 1019 / 8 = 127.375$.

(b) Sample Median: Average of 4th (127) and 5th (128) terms: $(127 + 128) / 2 = 127.5$.

(c) Mode: All observations occur exactly once → **No Mode**.

(d) Quartiles:

• Lower half = {120, 122, 125, 127} → $Q1 = (122 + 125) / 2 = 123.5$.

• Median = $Q2 = 127.5$.

• Upper half = {128, 130, 132, 135} → $Q3 = (130 + 132) / 2 = 131.0$.

(e) Interquartile Range (IQR): $IQR = Q3 - Q1 = 131.0 - 123.5 = 7.5$.

Question 2. Time-Series Analysis of Worldwide Fatal Airline Accidents

Fatal airline accidents worldwide from 1997 to 2005 are recorded as follows:

1997: 25 | 1998: 20 | 1999: 21 | 2000: 18 | 2001: 18 | 2002: 13 | 2003: 14 | 2004: 9 | 2005: 18.

Compute the mean, median, sample standard deviation s , and analyze the multi-year safety trend.

Step-by-Step Solution:

Sorted Dataset ($n = 9$): {9, 13, 14, 18, 18, 18, 20, 21, 25}.

- **Mean $\bar{x}$:** $\Sigma x = 156 \Rightarrow \bar{x} = 156 / 9 = 17.33 \text{ accidents/year}$.
- **Median:** 5th element in sorted order = **18.0**.
- **Sample Variance s^2 :** $\Sigma(x - \bar{x})^2 = 172.0 \Rightarrow s^2 = 172.0 / (9 - 1) = 21.50$.
- **Sample Standard Deviation s :** $s = \sqrt{21.50} \approx 4.64 \text{ accidents/year}$.
- **Trend Interpretation:** Annual fatal accidents declined significantly from 25 in 1997 to 9 in 2004, demonstrating progressive advancements in aviation safety protocols.

Question 3. 5-Number Summary & Outlier Detection

Find the 5-number summary and test for outliers using the $1.5 \times \text{IQR}$ rule for:

{10, 25, 28, 30, 32, 35, 38, 40, 42, 45, 75} ($n = 11$).

Step-by-Step Solution:

- **Minimum:** 10, **Maximum:** 75.
- **Median (Q2):** 6th term = **35**.
- **Q1:** Median of lower 5 values {10, 25, 28, 30, 32} = **28**.
- **Q3:** Median of upper 5 values {38, 40, 42, 45, 75} = **42**.
- **5-Number Summary:** [**10, 28, 35, 42, 75**].
- **Outlier Boundaries:** $\text{IQR} = 42 - 28 = 14$.
 Upper Boundary = $Q3 + 1.5(\text{IQR}) = 42 + 21 = 57$. Since $75 > 57$, **75 is an outlier**.
 Lower Boundary = $Q1 - 1.5(\text{IQR}) = 28 - 21 = 7$. Minimum $10 \geq 7$ (no lower outlier).

Question 4. z-Score Standardization & Relative Standing

A student scores 85 on a Biostatistics exam ($\mu = 75$, $\sigma = 8$) and 88 on a Chemistry exam ($\mu = 70$, $\sigma = 12$). Calculate both z-scores and determine in which subject the student scored higher relative to the class.

Step-by-Step Solution:

Formula: $z = (x - \mu) / \sigma$.

- **Biostatistics z-score:** $z = (85 - 75) / 8 = 10 / 8 = +1.25$.
- **Chemistry z-score:** $z = (88 - 70) / 12 = 18 / 12 = +1.50$.

Conclusion: The student performed relatively better in **Chemistry** (+1.50 SD above average) than in Biostatistics (+1.25 SD above average).

Question 5. Grouped Mean & Variance from Frequency Table

Given the frequency distribution of patient wait times (minutes):

• 0–4 min: $f = 5$ | 5–9 min: $f = 12$ | 10–14 min: $f = 8$ | 15–19 min: $f = 5$ (Total $n = 30$).

Compute the grouped mean $\bar{x}$ and sample variance s^2 .

Step-by-Step Solution:

Class Midpoints (x_m): 2, 7, 12, 17.

- **$\Sigma(f \cdot x_m)$:** $(5 \times 2) + (12 \times 7) + (8 \times 12) + (5 \times 17) = 10 + 84 + 96 + 85 = 275$.
- **Grouped Mean $\bar{x}$:** $275 / 30 = 9.167 \text{ minutes}$.
- **$\Sigma(f \cdot x_m^2)$:** $(5 \times 4) + (12 \times 49) + (8 \times 144) + (5 \times 289) = 20 + 588 + 1152 + 1445 = 3205$.
- **Sample Variance s^2 :** $s^2 = [3205 - (275)^2/30] / 29 = [3205 - 2520.833] / 29 = 684.167 / 29 \approx 23.59 \text{ min}^2$.

Question 6. Chebyshev's Theorem vs. Empirical Rule

For a distribution with $\mu = 100$ and $\sigma = 15$:

- Apply Chebyshev's theorem to find the minimum data proportion between 70 and 130.
- If the distribution is bell-shaped (normal), apply the Empirical Rule to find the percentage of data between 70 and 130.

Step-by-Step Solution:

The bounds $[70, 130]$ represent $100 \pm 2(15)$ $\Rightarrow k = 2$ standard deviations.

(a) Chebyshev's Theorem (Any distribution): Minimum proportion = $1 - 1/k^2 = 1 - 1/4 = 0.75$ (**at least 75%**).

(b) Empirical Rule (Bell-shaped normal distribution): Exactly **95%** of the dataset lies within $\mu \pm 2\sigma$ (70 to 130).

Question 7. Coefficient of Variation (CV) Comparative Analysis

Stock A has mean price \$50, SD = \$5. Stock B has mean price \$200, SD = \$10. Compute the Coefficient of Variation (CV = $(s / \bar{x}) \times 100\%$) for both stocks and compare their relative variability.

Step-by-Step Solution:

• **Stock A CV:** $(\$5 / \$50) \times 100\% = 10.0\%$.

• **Stock B CV:** $(\$10 / \$200) \times 100\% = 5.0\%$.

Interpretation: Stock A exhibits twice the relative volatility/variability (10.0%) compared to Stock B (5.0%), even though Stock B has a higher absolute dollar standard deviation (\$10 vs. \$5).